LEARNING MODULE

14

Random Variables, Expectation and Jensen's Inequality

A day's return as a random variable: its distribution, its expectation and moments, the tail bounds that need no model, and the inequality that is the real reason for volatility drag.

VOLUME III · Probability **BUILDS ON** LM1 · LM3 · LM4 · LM5 · LM13

LEADS TO LM15 · LM16 · LM17 · LM18 · LM19 · LM20 · LM21 · LM22 · LM26 · LM30 · LM32 · LM41

TIME ≈ 2.5 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	**1** describe a discrete or continuous random variable by its pmf or density and its distribution function, and find its quantiles and medians, including those of a sample's empirical distribution
☐	**2** calculate expectations as probability-weighted sums or integrals, check that they exist, and apply LOTUS and linearity, including to $g(f) = \mathbb{E}[\ln(1 + fR)]$
☐	**3** derive $\mathbb{E}[X^2] = \sigma_X^2 + \mu_X^2$, $\mathrm{Var}(\eta X + \zeta) = \eta^2 \mathrm{Var}(X)$, $\mathbb{E}[Z^4] = 3$ and $\mathrm{Var}(Z^2) = 2$, and compute the skewness and kurtosis of a distribution
☐	**4** apply Markov's, Chebyshev's and Cantelli's inequalities under their conditions, and compare their bounds with TAIEX's tail frequencies
☐	**5** prove Jensen's inequality and its strict form from the tangent-line property, and derive the weighted AM–GM inequality from it
☐	**6** explain volatility drag through the Jensen gap $\ln(1 + \mu) - g$ of the logarithm, compute both exactly, and judge when $\sigma^2/2$ describes them
☐	**7** compute the distribution, quantiles, moments and tail frequencies of an empirical distribution in Python, stating the quantile and moment conventions

LEARNING MODULE OVERVIEW

- A random variable is a function on a sample space. One distribution function $F_X(x) = \Pr(X \le x)$ serves discrete and continuous laws, and the quantile function $F_X^{-1}(u) = \min\{x : F_X(x) \ge u\}$ always exists, even where $F_X(x) = 1/2$ has no solution.
- As an empirical distribution, TAIEX's 5,849 daily total returns have a median of 0.0879%, above their mean of 0.0625%, and a longer lower tail: the 0.1% quantile is −6.313%, the 99.9% quantile 5.823%.
- An expectation is a probability-weighted sum or integral that exists only when it converges absolutely. LOTUS computes $\mathbb{E}[h(X)]$ without the law of $h(X)$, and the growth function $g(f) = \mathbb{E}[\ln(1 + fR)]$ is one such expectation.
- The identity $\mathbb{E}[X^2] = \sigma_X^2 + \mu_X^2$ is the population form of the sample identity (3.7). TAIEX's skewness is −0.34 and its kurtosis 8.27, an excess of 5.27; its fourth moment is 2.72 times that of the normal with the same mean and variance.
- Markov's, Chebyshev's and Cantelli's inequalities bound tail probabilities with a mean or a variance alone. They hold exactly for TAIEX's empirical law and are loose: 17 days lie at least four standard deviations below the mean, against Cantelli's bound of 344.1.
- Jensen's inequality, $\mathbb{E}[h(X)] \le h(\mathbb{E}X)$ for a concave h, follows from one tangent line. It gives the weighted AM–GM inequality and $R_G \le \mu$: the compound rate never exceeds the average return.
- The Jensen gap $\ln(1 + \mu) - g = \ln(1 + \mu) - \ln(1 + R_G)$ is volatility drag measured in log points. For TAIEX it is 0.7316 bp a day (the drag $\mu - R_G$ is 0.7320 bp), of which the second-order term gives 0.7292 bp; for annual returns the higher-order terms matter, and for a lognormal law the gap is exactly half the variance of the log return.

1 INTRODUCTION

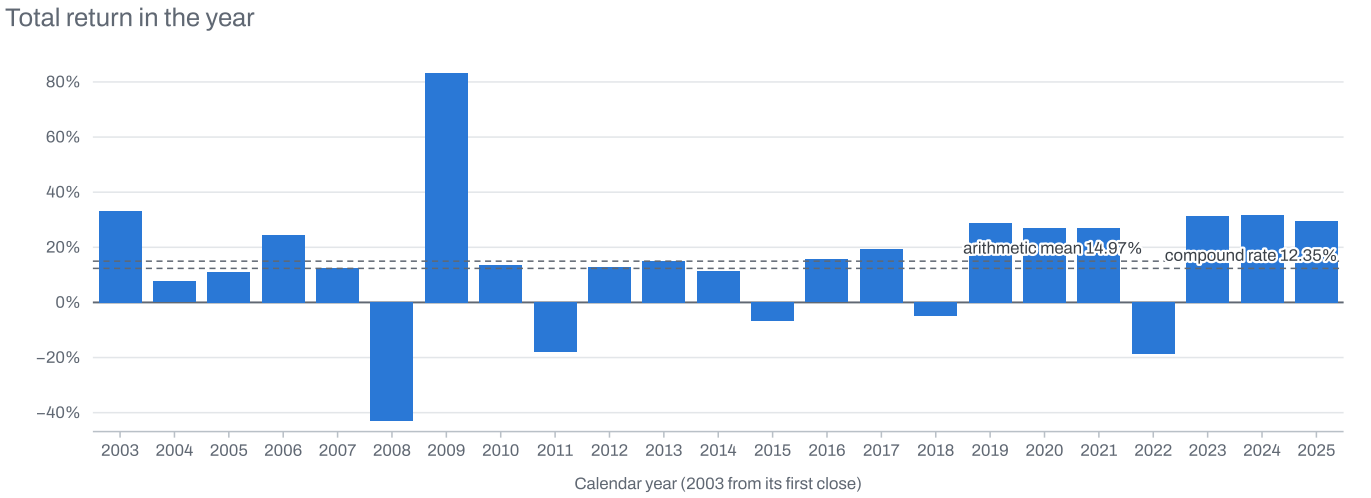
LM13 defined a random variable as a function on a sample space. This module computes with random variables. Its central result, Jensen's inequality, compares the average of a function with the function of the average. For the logarithm, the gap between the two is the Jensen gap, LM3's volatility drag measured in log points.

MOTIVATING CASE **Two averages, one market**

The TAIEX total-return index reinvests cash dividends. Over the 23 calendar years 2003–2025 (2003 from its first close, 2 January 2003), it returned 14.97% a year on average. An investor who held it throughout earned 12.35% a year. The average of 2008's −43.07% and 2009's +83.34% is +20.1% a year, yet together the two years returned only +4.4%.

Day by day, through 8 October 2026, the average return was 0.0625%, while the compound daily rate was 0.0552%. Their gap, the volatility drag of LM3, is 0.7320 bp, within 0.002 bp of half the daily variance, 0.7301 bp. Why is the compound rate never above the average? What exactly is the drag? And when does $\sigma^2/2$ describe it?

Exhibit 1: TAIEX's calendar-year total returns, 2003–2025



Note: dashed lines at the arithmetic mean of the 23 returns, 14.97% (upper), and the compound rate, 12.35% (lower). The extremes are 2008 (−43.07%) and 2009 (+83.34%). 2003 runs from its first close; 2026 (partial) is excluded. The case's daily inputs, to six decimals of a percent, are $\bar{R} = 0.062505\%$, $R_G = 0.055185\%$ and $s = 1.208364\%$ (four decimals give only 0.73 bp). Source: FinMind `TaiwanStockTotalReturnIndex` (data_id TAIEX), retrieved 18 October 2026; dividends reinvested, no costs. Computations: `data/lm14_numbers.py` and `LM14_lab.ipynb`.

Sections 2 to 4 cover distributions, expectations and moments, and Section 5 tail bounds. Section 6 proves Jensen's inequality, Section 7 resolves the case, and Section 8 works in Python.

NOTE **Notation for this module**

TAIEX is the total-return index, with daily simple return R and log return $r = \ln(1+R)$: $T = 5{,}849$ returns (a count) from 3 January 2003 to 8 October 2026; the closes from 2 January 2003 span 23.7645 years, so $A = 246.12$. Calendar years are 2003–2025, 2003 from its first close.

- $\hat{F}$, $\hat{F}^{-1}$, $\mathbb{E}_{\hat{F}}$: the empirical distribution of a sample, its quantile function and its expectation. μ, σ, $g = \mathbb{E}[\ln(1+R)]$ and the compound rate $R_G := e^g - 1$ belong to R; under $\hat{F}$ they are $\bar{R}$, s, $\bar{r}$ (divisor T) and LM3's geometric mean return R_G. s_r: the standard deviation of r.
- Local: x_j, J: the values of a discrete X and their number (J may be infinite), or in Section 6 the numbers of the AM–GM inequality, with weights w_j; $R_{(j)}$: the return of rank j from the smallest; h: a generic function on an open interval $\mathcal{I}$ (LM4's I is now the identity matrix). Indices: j for values and ranks, t for dates.
- The volatility drag $\mu - R_G$ and the Jensen gap $\ln(1+\mu) - g$ have no symbol.

NOTE **Research hygiene**

Every TAIEX frequency, quantile and moment here describes the sample through its empirical distribution: these numbers are **descriptive**, not forecasts. The normal model with the same mean and variance is **in-sample**. The coins, dice, two-point bets and the lognormal form are **models**, and the lab's comparison of the two price-limit regimes is a **stability check**. LM17 asks when averages approach expectations, and LM20 treats estimation.

2 RANDOM VARIABLES AND THEIR DISTRIBUTIONS

- ☐ describe a discrete or continuous random variable by its pmf or density and its distribution function, and find its quantiles and medians, including those of a sample's empirical distribution

A random variable assigns a number to every outcome of a sample space Ω (LM13). Several outcomes can share a value, and the **distribution** of X, also called its **law**, collects their probabilities.

EXAMPLE 1

Two trades as one random variable

LM4's bet returns +2% or −1% with equal probability; at its Kelly exposure $f = 25$, one trade multiplies wealth by 1.5 or 0.75. Two independent trades give LM13's sample space $\Omega = \{++, +-, -+, --\}$, each outcome with probability ¼. LM13 tabulated the distribution of $X = W_2/W_0$. Find its distribution function, its 10% quantile and its median.

SOLUTION

The outcomes map to 2.25, 1.125, 1.125 and 0.5625 (Exhibit 2), so $p_X(1.125) = \frac{1}{2}$ and $p_X(0.5625) = p_X(2.25) = \frac{1}{4}$.

The distribution function jumps from ¼ to ¾ at 1.125, so $F_X(x) = \frac{1}{2}$ has no solution. Yet 1.125 is the only median under the definition below, since $\Pr(X \le 1.125) = \Pr(X \ge 1.125) = \frac{3}{4}$. The 10% quantile, the smallest x with $F_X(x) \ge 0.1$, is 0.5625.

Exhibit 2: Two trades of the +2%/−1% bet at $f = 25$

Outcome ω	$\Pr(\omega)$	$X(\omega)$	$\frac{1}{2}\ln X(\omega)$
$++$	¼	2.25	40.5465%
$+-$	¼	1.125	5.8892%
$-+$	¼	1.125	5.8892%
$--$	¼	0.5625	−28.7682%

Value x	$p_X(x)$	$F_X(x)$
0.5625	¼	¼
1.125	½	¾
2.25	¼	1

Note: $X = W_2/W_0$ is the growth factor of two independent trades. Exact values; logs to four decimals of a percent.

Distribution functions and quantiles

A discrete random variable has a **probability mass function** (pmf) $p_X(x_j) := \Pr(X = x_j)$ on its finitely or countably many values x_j (LM13). A continuous one has a **density** f_X (LM5).

DEFINITION **Distribution function, quantile function and median**

The **distribution function** of a random variable X is $F_X(x) := \Pr(X \le x)$, (13.11). For a discrete X it is the sum of $p_X(x_j)$ over all $x_j \le x$; for a continuous one, the area under f_X to the left of x. For $0 < u < 1$, the **quantile function** is

$$F_X^{-1}(u) := \min\{x : F_X(x) \ge u\}.$$

A value x is a **median** of X if $\Pr(X \le x) \ge \frac{1}{2}$ and $\Pr(X \ge x) \ge \frac{1}{2}$.

The minimum exists because F_X rises from 0 to 1, never decreases and takes the upper value at a jump. When F_X is continuous and strictly increasing, F_X^{-1} is its inverse and a median solves $F_X(x) = \frac{1}{2}$, as in LM5. In general the medians form a closed interval, possibly a single point.

A sample as a random variable

A sample becomes a random variable through its **empirical distribution** $\hat{F}$, (13.12), with mass $1/T$ on each of TAIEX's T returns, so every frequency is a probability. From this volume on, **population** names a property of a random variable or a model (μ, σ), and **sample** a statistic of data ($\bar{R}$, s, divisor T). LM3's "population variance" s^2 is the variance of $\hat{F}$.

Quantiles of $\hat{F}$ are **order statistics**. Sort the returns, which are all distinct, as $R_{(1)} < R_{(2)} < \cdots < R_{(T)}$. Because $\hat{F}(R_{(j)}) = j/T$, the smallest order statistic with $j/T \geq u$ is

$$\hat{F}^{-1}(u) = R_{(\lceil uT \rceil)}.$$

EXAMPLE 2

TAIEX's daily return as a random variable

Treat TAIEX's daily returns as a random variable R under $\hat{F}$. Find $\hat{F}(0)$, the median and the 1% and 99% quantiles.

SOLUTION

No day returned exactly 0, and 2,644 days fell, so $\hat{F}(0) = 2{,}644/5{,}849 = 0.4520$, the value of LM13; the other 3,205 days (54.8%) rose. Because T is odd, the median is the 2,925th return, 0.0879%, above the mean of 0.0625%: 3,020 days (51.6%) beat the mean.

The lower tail reaches further (Exhibit 3). The 1% quantile is $\hat{F}^{-1}(0.01) = -3.567\%$, against $\hat{F}^{-1}(0.99) = 3.125\%$; at 0.1% it is −6.313%, against 5.823% at 99.9%.

Exhibit 3: Quantiles of TAIEX's daily returns

u	$\lceil uT \rceil$	$\hat{F}^{-1}(u)$ (%)	`np.quantile` default (%)
0.001	6	−6.313	−5.866
0.01	59	−3.567	−3.567
0.05	293	−1.862	−1.857
0.25	1,463	−0.490	−0.490
0.5	2,925	0.088	0.088
0.75	4,387	0.681	0.681
0.95	5,557	1.833	1.832
0.99	5,791	3.125	3.120
0.999	5,844	5.823	5.666

Note: TAIEX's 5,849 daily total returns. NumPy's default, `method="linear"`, interpolates between order statistics at position $(T-1)u$. The columns agree exactly where $(T-1)u$ is whole (the quartiles and the median). Elsewhere they differ, at $u = 0.01$ only in the fourth decimal, and most in the far tails. Source: as Exhibit 1.

LM32 turns lower quantiles into value at risk; LM15 compares them with a normal model's.

PITFALL **A median need not solve $F_X(x) = \frac{1}{2}$**

A fair coin paying 0 or 1 has $F_X(x) = \frac{1}{2}$ on all of $[0, 1)$, and every x in $[0, 1]$ is a median. Example 1 has one median, although $F_X(x) = \frac{1}{2}$ has no solution. Software reports one point by its own rule (Section 8), so state the definition.

KNOWLEDGE CHECK 1

A fair die shows 1 to 6, each with probability 1/6. Which values are medians, and what is $F_X^{-1}(\frac{1}{2})$?

ANSWER

Every x in $[3, 4]$ is a median, since then $\Pr(X \le x) \ge \frac{3}{6}$ and $\Pr(X \ge x) \ge \frac{3}{6}$. The quantile function picks the smallest, $F_X^{-1}(\frac{1}{2}) = 3$, while `np.median` of the six faces returns the midpoint 3.5.

3 EXPECTATION AND LOTUS

- [ ] calculate expectations as probability-weighted sums or integrals, check that they exist, and apply LOTUS and linearity, including to $g(f) = \mathbb{E}[\ln(1 + fR)]$

An expectation is a probability-weighted average of values.

DEFINITION **Expectation**

Let X take the values x_j, $j = 1, \dots, J$ (J finite or infinite), with pmf p_X. If $\sum_j |x_j|\, p_X(x_j) < \infty$, the **expectation** $\mathbb{E}[X]$ is the sum below; otherwise it does not exist. The **indicator** $\mathbf{1}_{\mathcal{A}}$ of an event $\mathcal{A}$ is 1 on $\mathcal{A}$ and 0 elsewhere, so

$$\begin{aligned} \mathbb{E}[X] &:= \sum_j x_j\, p_X(x_j), \\ \mathbb{E}[\mathbf{1}_{\mathcal{A}}] &= \Pr(\mathcal{A}). \end{aligned} \tag{14.1}$$

For a density, LM5's (5.9) is the continuous form, with the same condition. Random variables that are neither discrete nor continuous need a more general integral, which this book does not use. Under $\hat{F}$, (14.1) gives $\mathbb{E}_{\hat{F}}[R] = \bar{R}$.

An expectation can fail to exist: LM5's Cauchy density has no mean, and Problem 3 has a discrete case. Why the average of many independent draws approaches the expectation is the law of large numbers of LM17.

THEOREM **Law of the unconscious statistician (LOTUS)**

Let X be discrete with values x_j, and let h be a function with $\sum_j |h(x_j)|\, p_X(x_j) < \infty$. Then

$$\mathbb{E}[h(X)] = \sum_j h(x_j)\, p_X(x_j). \tag{14.2}$$

LOTUS weights each value of X by its probability and never needs the pmf of $h(X)$. For densities, (5.9) is LOTUS: when $Y = h(X)$ has a density, (5.9) equals $\int y\, f_Y(y)\, dy$, for a continuous h and an integral that exists. This is stated without proof (Grinstead and Snell 1997, §6.3, Theorem 6.11).

DEEP DIVE **LOTUS by grouping** optional · click to expand

Here y_i, $i = 1, 2, \ldots$, are the distinct values of $Y = h(X)$. Countable additivity (LM13) gives $\Pr(Y = y_i) = \sum_{j:\, h(x_j) = y_i} p_X(x_j)$. Grouping the terms of (14.2) by the value of h,

$$\begin{aligned} &\sum_j h(x_j)\, p_X(x_j) \\ &\quad = \sum_i \sum_{j:\, h(x_j) = y_i} y_i\, p_X(x_j) \\ &\quad = \sum_i y_i \Pr(Y = y_i), \end{aligned}$$

which is $\mathbb{E}[Y]$ by (14.1). Regrouping a series is allowed when it converges absolutely, the condition of (14.2), and the same regrouping of the absolute values shows that $\mathbb{E}[Y]$ exists. ■

PROPOSITION **Linearity**

Let X and Y be random variables on one sample space whose expectations exist, and let η and ζ be constants. Then

$$\begin{aligned} \mathbb{E}[\eta X + \zeta] &= \eta\, \mathbb{E}[X] + \zeta, \\ \mathbb{E}[X + Y] &= \mathbb{E}[X] + \mathbb{E}[Y]. \end{aligned} \tag{14.3}$$

On a countable Ω, grouping the outcomes by the value of X gives $\mathbb{E}[X] = \sum_\omega X(\omega) \Pr(\omega)$, and both identities follow by splitting one absolutely convergent sum into two. For densities, $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$ is stated without proof (Grinstead and Snell 1997, §6.3, Theorem 6.10). It follows that $\mathbb{E}[X] \le \mathbb{E}[Y]$ when $X \le Y$ with probability one, since outcomes of probability zero add nothing to the sums.

EXAMPLE 3

Expectation two ways

For the X of Example 1, compute $\mathbb{E}[X]$ over the outcomes and over the values. Then compare $\mathbb{E}[\frac{1}{2} \ln X]$, by LOTUS, with $\frac{1}{2} \ln \mathbb{E}[X]$.

SOLUTION

Over the outcomes, $\frac{1}{4}(2.25 + 1.125 + 1.125 + 0.5625) = 1.265625$, and over the values $\frac{1}{4} \times 0.5625 + \frac{1}{2} \times 1.125 + \frac{1}{4} \times 2.25 = 1.265625$. That is 1.125^2, the product of the two trades' expected factors $1 + 25 \times 0.5\%$. The expectation of a product of independent factors is the product of the expectations, a rule that LM16 proves. Here the four outcomes give it directly.

LOTUS with $h(x) = \frac{1}{2}\ln x$, the log growth per trade, uses Exhibit 2's last column:

$$\begin{aligned}&\tfrac{1}{4}(40.5465\%) + \tfrac{1}{2}(5.8892\%)\\ &\quad + \tfrac{1}{4}(-28.7682\%) = 5.889\%.\end{aligned}$$

This average is $\frac{1}{2}\ln 1.125$, LM4's g^* again. Against it, $\frac{1}{2}\ln \mathbb{E}[X] = \ln 1.125 = 11.778\%$: here the log of the average is twice the average of the logs.

EXAMPLE 4

The growth function as an expectation

Show that LM4's growth function $g(f) = \mathbb{E}[\ln(1 + fR)]$, (4.1), is LOTUS, and evaluate it for TAIEX at $f = 1$.

SOLUTION

With $h(x) = \ln(1 + fx)$ and $R \in \{b, -a\}$, (14.2) gives (4.7), $p\ln(1 + fb) + q\ln(1 - fa)$. Under $\hat{F}$ it gives (4.4), $\bar{g}(f) = \mathbb{E}_{\hat{F}}[\ln(1 + fR)]$.

At $f = 1$, $\bar{g}(1) = \bar{r} = 5.5170$ bp a day. Times A, this is 13.58% a year, which equals $\ln(25.2007)/23.7645$ by (1.12): the log growth over the whole sample, 2026 included. The compound rate is $25.2007^{1/23.7645} - 1 = 14.54\%$.

Differentiating under the expectation, as (4.5) does term by term, needs a condition when R takes infinitely many values; LM41 states it.

PITFALL **The expectation of a function is not the function of the expectation**

For TAIEX, $g = \bar{r}$ is not $\ln(1 + \bar{R})$, which is larger by 0.7316 bp a day (Section 7), and Example 3 shows a factor of two. By (14.3) the two agree for every X when $h(x) = \eta x + \zeta$; two-point laws show that no other h does so.

導讀 **隨機變數、分位數與期望值**

隨機變數（random variable）是樣本空間（sample space）上的函數：離散時用機率質量函數（probability mass function），連續時用密度（density），兩者共用同一個累積分布函數（distribution function）$F_X(x) = \Pr(X \le x)$。分位數函數（quantile function）取「使 $F_X(x) \ge u$ 的最小 x」，所以一定存在；中位數（median）卻可能是一整段區間，也可能根本解不出 $F_X(x) = \frac{1}{2}$。把加權股價報酬指數 5,849 個日報酬各給 $1/T$ 的機率，就得到經驗分布（empirical distribution），樣本的頻率就是這個模型的機率。樣本平均數（sample mean）就是經驗分布的期望值（expectation）；LM3 所說的母體變異數（population variance）s^2，也就是經驗分布的變異數。LOTUS 讓我們不必先求 $h(X)$ 的分布（distribution），直接用 X 的機率加權 $h(x)$，但函數的期望值一般不等於期望值的函數，$g = \mathbb{E}[\ln(1+R)]$ 並不等於 $\ln(1+\mu)$。

KNOWLEDGE CHECK 2

A day returns +10% or −10% with equal probability. Compute $\mathbb{E}[R]$ and $\mathbb{E}[\ln(1+R)]$.

ANSWER

The mean return is 0, but $\mathbb{E}[\ln(1+R)] = \frac{1}{2}\ln 1.1 + \frac{1}{2}\ln 0.9 = \frac{1}{2}\ln 0.99 = -0.5025\%$. The log of the average, $\ln 1 = 0$, is not the average of the logs.

4 VARIANCE AND HIGHER MOMENTS

- ☐ derive $\mathbb{E}[X^2] = \sigma_X^2 + \mu_X^2$, $\mathrm{Var}(\eta X + \zeta) = \eta^2 \mathrm{Var}(X)$, $\mathbb{E}[Z^4] = 3$ and $\mathrm{Var}(Z^2) = 2$, and compute the skewness and kurtosis of a distribution

The **variance** of X is $\mathrm{Var}(X) := \mathbb{E}[(X - \mu_X)^2]$, where $\mu_X = \mathbb{E}[X]$, and $\sigma_X := \sqrt{\mathrm{Var}(X)}$ is its **standard deviation**.

PROPOSITION **Second moment and variance**

If $\mathbb{E}[X^2] < \infty$, then $\mathbb{E}[X]$ exists, and for constants η and ζ

$$\begin{aligned} \mathbb{E}[X^2] &= \sigma_X^2 + \mu_X^2, \\ \mathrm{Var}(\eta X + \zeta) &= \eta^2 \mathrm{Var}(X). \end{aligned} \tag{14.4}$$

The mean exists because $|x| \le 1 + x^2$ for every x. Expanding the square with (14.3) gives $\mathbb{E}[(X - \mu_X)^2] = \mathbb{E}[X^2] - 2\mu_X\mathbb{E}[X] + \mu_X^2 = \mathbb{E}[X^2] - \mu_X^2$. For the second identity, $\eta X + \zeta$ has mean $\eta\mu_X + \zeta$, so its deviation from the mean is $\eta(X - \mu_X)$.

Under $\hat{F}$, (14.4) is LM3's identity (3.7), $\frac{1}{T}\sum_t R_t^2 = s^2 + \bar{R}^2$; in a model it is LM4's $\mathbb{E}[R^2] = \sigma^2 + \mu^2$. The variance of a sum needs the covariance of its terms (LM16). By (14.4), $(X - \mu_X)/\sigma_X$ has mean 0 and variance 1, as LM5's (5.5) used for the normal. Since $x^2 \le 1 + x^4$ and $|x|^3 \le 1 + x^4$, a finite fourth moment makes every lower moment finite.

DEFINITION **Skewness and kurtosis**

Let $\mathbb{E}[X^4] < \infty$ and $\sigma_X > 0$. The **skewness** and the **kurtosis** of X are its third and fourth standardized moments,

$$\begin{aligned} \operatorname{Skew}(X) &:= \frac{\mathbb{E}[(X - \mu_X)^3]}{\sigma_X^3}, \\ \operatorname{Kurt}(X) &:= \frac{\mathbb{E}[(X - \mu_X)^4]}{\sigma_X^4}, \end{aligned} \tag{14.5}$$

and the **excess kurtosis** is $\operatorname{Kurt}(X) - 3$. The skewness needs only $\mathbb{E}|X|^3 < \infty$.

Both are unchanged when X becomes $\eta X + \zeta$ with $\eta > 0$, because the deviation and σ_X scale by the same η. Applied to a sample, (14.5) gives the sample skewness and kurtosis. The choice of T or $T - 1$, and how far such numbers move between samples, are questions for LM19 and LM20.

EXAMPLE 5

Moments of the +2%/−1% bet

Compute the moments of LM4's bet, +2% or −1% with equal probability, up to the kurtosis.

SOLUTION

The mean is $\mu = 0.5\%$, and the deviations are ±1.5%, so $\sigma = 1.5\%$ and $\mathbb{E}[R^2] = \frac{1}{2}(0.02^2 + 0.01^2) = 0.00025 = \sigma^2 + \mu^2$. The raw third moment is $\frac{1}{2}(0.02^3 - 0.01^3) = 3.5 \times 10^{-6}$, which equals $\mu^3 + 3\mu\sigma^2$, as in LM4's Practice Problem 13.

The deviations are symmetric, so the skewness is 0, although the raw third moment is positive. Every squared deviation equals σ^2, so the kurtosis is 1, an excess kurtosis of −2.

The normal's moments

For a standard normal Z the odd moments vanish by symmetry, and $\mathbb{E}[Z^2] = 1$ (LM5). Integration by parts, (5.6), moves one factor z onto φ, since $z\varphi(z)$ is the derivative of $-\varphi(z)$:

$$\begin{aligned}\mathbb{E}[Z^4] &= \int_{-\infty}^{\infty} z^3 \cdot z\varphi(z)\,dz \\ &= \big[-z^3\varphi(z)\big]_{-\infty}^{\infty} + 3\,\mathbb{E}[Z^2] = 3.\end{aligned}$$

Hence $\operatorname{Var}(Z^2) = \mathbb{E}[Z^4] - (\mathbb{E}[Z^2])^2 = 2$, the value LM3 used without proof. For a normal $X = \mu + \sigma Z$, expanding the powers of $\mu + \sigma Z$ and dropping the odd moments of Z gives

$$\begin{gathered}\mathbb{E}[Z^3] = 0, \quad \mathbb{E}[Z^4] = 3, \quad \operatorname{Var}(Z^2) = 2, \\ \mathbb{E}[X^3] = \mu^3 + 3\mu\sigma^2, \\ \mathbb{E}[X^4] = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4.\end{gathered} \tag{14.6}$$

So the normal has skewness 0 and kurtosis 3, which LM5 stated without proof. Excess kurtosis measures tail weight against this benchmark.

EXAMPLE 6

TAIEX against the normal

Compare TAIEX's moments under $\hat{F}$ with those of $\mathcal{N}(\bar{R}, s^2)$. Then size the terms that LM4's normal approximations $\mathbb{E}[R^3] \approx 3\mu\sigma^2$ and $\mathbb{E}[R^4] \approx 3\sigma^4$ drop.

SOLUTION

Under $\hat{F}$, (14.5) gives a skewness of −0.34 and a kurtosis of 8.27, an excess kurtosis of 5.27 (Exhibit 4). The raw third moment is -3.231×10^{-7}, against the normal's $+2.740 \times 10^{-7}$ from (14.6). The fourth moment, 1.751×10^{-7}, is 2.72 times the normal's 6.430×10^{-8}; LM4 found 3.7 for 0050, another sample. LM15 fits distributions to TAIEX; this normal is only a benchmark.

By (14.6), the approximations drop terms of relative size $(\bar{R}/s)^2 = 0.0027$: a share $\frac{1}{3}(\bar{R}/s)^2 = 0.09\%$ of $3\bar{R}s^2 = 2.738 \times 10^{-7}$, and a share $2(\bar{R}/s)^2 + \frac{1}{3}(\bar{R}/s)^4 = 0.54\%$ of $3s^4 = 6.396 \times 10^{-8}$.

Exhibit 4: Moments of TAIEX's daily returns and of the normal with the same mean and variance

Moment	TAIEX, $\hat{F}$	Normal, $\mathcal{N}(\bar{R}, s^2)$
Mean	0.0625%	0.0625%
Standard deviation	1.2084%	1.2084%
$\mathbb{E}[R^2]$	1.464×10^{-4}	1.464×10^{-4}
$s^2 + \bar{R}^2$, (3.7)	$1.460 \times 10^{-4} + 3.907 \times 10^{-7}$	$1.460 \times 10^{-4} + 3.907 \times 10^{-7}$
$\mathbb{E}[R^3]$	-3.231×10^{-7}	2.740×10^{-7}
$\mathbb{E}[R^4]$	1.751×10^{-7}	6.430×10^{-8}
Skewness	−0.338	0
Kurtosis	8.27	3
Excess kurtosis	5.27	0
$\mathbb{E}[R^4]$, ratio to the normal	2.72	1

Note: divisor T; raw moments to four significant figures, which add up in the (3.7) row. The normal's moments use (14.6) and the six-decimal $\bar{R}$ and s of Exhibit 1's note. 0050's excess kurtosis of 8.12 (LM4, LM5) belongs to another sample. Source: as Exhibit 1.

PITFALL **A raw third moment is not skewness, and kurtosis is not excess kurtosis**

Expanding $\mathbb{E}[(\mu + (R - \mu))^3]$ gives $\mathbb{E}[R^3] = \mu^3 + 3\mu\sigma^2 + \mathbb{E}[(R - \mu)^3]$: the raw moment mixes the mean into the asymmetry, so Example 5's bet has $\mathbb{E}[R^3] > 0$ and zero skewness. Kurtosis and excess kurtosis differ by 3: TAIEX's 8.27 is 5.27 in excess, the number to set beside 0050's 8.12. Say "excess" whenever such a number meets LM4, LM5 or LM15.

導讀 **動差：分布形狀的摘要**

動差（moment）是 X 各次方的期望值（expectation）；變異數（variance）是二階中心動差（central moment），而 $\mathbb{E}[X^2] = \sigma_X^2 + \mu_X^2$ 正是 LM3 恆等式 (3.7) 的母體（population）版本。把 X 減去平均數（mean）、除以標準差（standard deviation）以後再取三次方與四次方的期望值，就是標準化動差（standardized moment），分別稱為偏態（skewness）與峰態（kurtosis）。常態分布（normal distribution）的峰態是 3，所以峰態減 3 叫超額峰態（excess kurtosis）。加權股價報酬指數日報酬的偏態是 −0.34、峰態 8.27，四階動差是同平均數、同變異數常態的 2.72 倍，大波動比常態多。原始三階動差（raw third moment）不是偏態，因為它混入了平均數：+2%/−1% 賭局的三階動差為正，偏態卻是 0。

KNOWLEDGE CHECK 3

A fund returns $2R$, twice TAIEX's daily return. What are its standard deviation and $\mathbb{E}[(2R)^2]$, and do its skewness and kurtosis change?

ANSWER

By (14.4) the standard deviation doubles, $2s = 2.417\%$; four decimals would need s to five, since $2 \times 1.2084 = 2.4168$ while $2s = 2.41673\%$. The second moment quadruples, $4\mathbb{E}[R^2] = 5.856 \times 10^{-4}$. Skewness and kurtosis do not change, because $\eta = 2 > 0$.

5 TAIL BOUNDS: MARKOV, CHEBYSHEV AND CANTELLI

- ☐ apply Markov's, Chebyshev's and Cantelli's inequalities under their conditions, and compare their bounds with TAIEX's tail frequencies

How often can a random variable stray far from its mean, if only its mean or its variance is known? Three inequalities answer without a model.

THEOREM Markov's inequality

If $X \geq 0$ with probability one and $x > 0$, then

$$\Pr(X \geq x) \leq \frac{\mathbb{E}[X]}{x}. \tag{14.7}$$

Proof. With probability one, $x\mathbf{1}\{X \geq x\} \leq X$: the left side is x when $X \geq x$ and 0 otherwise. Taking expectations with (14.1) and monotonicity gives $x \Pr(X \geq x) \leq \mathbb{E}[X]$. ■

THEOREM Chebyshev's and Cantelli's inequalities

Let X have a finite variance with $\sigma_X > 0$. For $k > 0$,

$$\Pr(|X - \mu_X| \geq k\sigma_X) \leq \frac{1}{k^2}, \tag{14.8}$$

and for $k \geq 0$ the lower tail alone satisfies

$$\Pr(X - \mu_X \leq -k\sigma_X) \leq \frac{1}{1 + k^2}. \tag{14.9}$$

Chebyshev's (14.8) is Markov's inequality for $(X - \mu_X)^2$ at the level $k^2\sigma_X^2$, since $\mathbb{E}[(X - \mu_X)^2] = \sigma_X^2$. Cantelli's one-sided (14.9) needs a shift (Deep Dive below). Simple laws attain both bounds, so neither

can be improved for every law.

For $k \le 1$ Chebyshev's bound says nothing. Without a finite variance, as for the Pareto law of Problem 7, neither bound applies. Under $\hat{F}$, with $\mu = \bar{R}$ and $\sigma = s$ (divisor T), every bound holds exactly, because $\hat{F}$ is a probability law with these moments.

DEEP DIVE **Cantelli's inequality, and the laws that attain the bounds** optional · click to expand

Here $t > 0$ is a shift, not a date. If $X - \mu_X \le -k\sigma_X$, then $\mu_X - X + t \ge k\sigma_X + t > 0$, so

$$\{X - \mu_X \le -k\sigma_X\} \subseteq \{(\mu_X - X + t)^2 \ge (k\sigma_X + t)^2\}.$$

The variable $\mu_X - X + t$ has mean t and variance σ_X^2, so its square has expectation $\sigma_X^2 + t^2$ by (14.4). Markov's (14.7) then gives

$$\Pr(X - \mu_X \le -k\sigma_X) \le \frac{\sigma_X^2 + t^2}{(k\sigma_X + t)^2}.$$

For $k > 0$ the derivative in t vanishes where $tk\sigma_X = \sigma_X^2$, so the bound is smallest at $t = \sigma_X/k$, where it equals $1/(1+k^2)$. At $k = 0$ the bound 1 is trivial.

The law with $\mu_X - k\sigma_X$ with probability $1/(1+k^2)$ and $\mu_X + \sigma_X/k$ otherwise has mean μ_X and variance σ_X^2, and it attains (14.9). For $k \ge 1$, the law with $\mu_X \pm k\sigma_X$ with probability $1/(2k^2)$ each and μ_X otherwise attains (14.8). ■

EXAMPLE 7

Three bounds for a 3% day

Bound the share of TAIEX days with $|R| \ge 3\%$ by Markov's inequality on $|R|$ and on R^2, and the share with $R \le -3\%$ by Cantelli's. Use $\mathbb{E}|R| = 0.8402\%$, $\mathbb{E}[R^2] = 1.464 \times 10^{-4}$, $\bar{R} = 0.0625\%$ and $s = 1.2084\%$.

SOLUTION

Markov's inequality on $|R|$ gives $\Pr(|R| \ge 3\%) \le 0.8402/3 = 28.0\%$. On R^2 the event is $\{R^2 \ge 0.03^2\}$, with the bound $1.464 \times 10^{-4}/0.0009 = 16.27\%$, tighter because R^2 weights large moves more. The data have 175 such days, 2.99%.

For the lower tail, $-3\% = \bar{R} - ks$ with $k = (0.0625 + 3)/1.2084 = 2.534$. Cantelli's bound is $1/(1 + 2.534^2) = 13.47\%$; the data have 108 days at or below −3% (LM13), 1.85%.

EXAMPLE 8

Four standard deviations below the mean

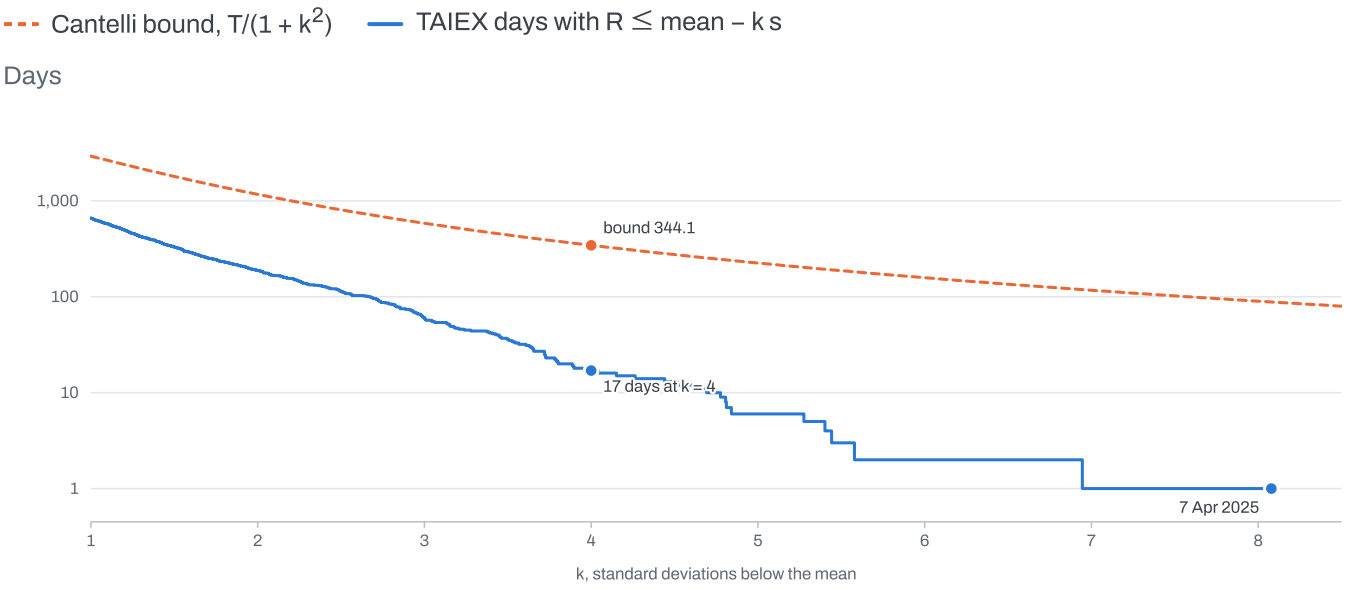
LM5's Practice Problem 9 set a normal four-sigma probability beside Cantelli's bound. Repeat it on TAIEX: how many days fell at least four standard deviations below the mean?

SOLUTION

The threshold is $\bar{R} - 4s = -4.771\%$. Cantelli's bound is $1/17 = 5.88\%$ of the days, and the normal model gives $\Phi(-4) = 3.17 \times 10^{-5}$, or 0.185 expected days in 5,849. TAIEX had 17 such days (0.29%): 92 times the normal's expected count and about a twentieth of the bound. These ratios are descriptive, not a test.

Two-sided, 30 days (0.51%) moved at least four standard deviations either way (Exhibit 6). Chebyshev's bound is 6.25%, and the normal model expects 0.37 days.

Exhibit 5: TAIEX's lower tail against Cantelli's bound



Note: days with $R \le \bar{R} - ks$ (step line) and Cantelli's bound in days, $T/(1+k^2)$: 2,924.5 at $k = 1$, 344.1 at $k = 4$ and 90.0 at $k = 8$; log scale. The step ends at $k = 8.08$, 7 April 2025 (−9.6997%). The daily price limit was 7% until 29 May 2015 and 10% from 1 June 2015 (LM5). On 7 April 2025 many stocks closed limit-down, which capped that day's fall. Source: as Exhibit 1.

Exhibit 6: TAIEX's tail days against Chebyshev's and Cantelli's bounds

k	Days $\lvert R-\bar{R}\rvert \geq ks$	Share	Chebyshev $1/k^2$	Days $R-\bar{R} \leq -ks$	Share	Cantelli $1/(1+k^2)$
1	1,313	22.45%	100.00%	665	11.37%	50.00%
2	319	5.45%	25.00%	188	3.21%	20.00%
3	99	1.69%	11.11%	61	1.04%	10.00%
4	30	0.51%	6.25%	17	0.29%	5.88%
5	11	0.19%	4.00%	6	0.10%	3.85%
6	4	0.07%	2.78%	2	0.03%	2.70%

Note: TAIEX's 5,849 daily total returns with $\bar{R}$ and s (divisor T); shares of T, counted with the guard of Section 8. LM15 compares the tails with a normal model's. Source: as Exhibit 1.

PITFALL **A bound is not a model**

The bounds hold for every law with the given variance, even the simple laws that attain them. So they lie far above real frequencies: 17 days against 344.1 at $k = 4$ (Exhibit 5). Read neither a bound nor the normal's 0.185 days as an estimate of risk; LM15 fits models to the tails.

導讀 **只知道平均數與變異數時**

馬可夫不等式（Markov's inequality）只用到 $X \geq 0$ 與平均數（mean）：門檻乘上指示函數（indicator）永遠不超過 X 本身，取期望值（expectation）就得到上界（bound）。對 $(X-\mu_X)^2$ 用同一招，就是謝比雪夫不等式（Chebyshev's inequality），上界是 $1/k^2$。加上一個平移再用一次，得到單邊的 Cantelli 不等式（Cantelli's inequality），上界是 $1/(1+k^2)$。這些上界對所有變異數（variance）有限的分布（distribution）都成立，所以對加權股價報酬指數的經驗分布（empirical distribution）也一定成立，但很鬆：低於平均數至少四個標準差（standard deviation）的上界是 344.1 天，實際只有 17 天。上界不是模型，常態模型算出的 0.185 天也不是風險估計；尾部模型留給 LM15。

KNOWLEDGE CHECK 4

What does Chebyshev's inequality say at $k = 1$? At $k = 2$, compare its two-sided bound with Cantelli's one-sided bound and with TAIEX.

ANSWER

At $k = 1$ the bound is 1, which says nothing. At $k = 2$, Chebyshev allows 25% of the days two standard deviations away on either side, and Cantelli 20% on the lower side alone. TAIEX had 319 days (5.45%) and 188 days (3.21%).

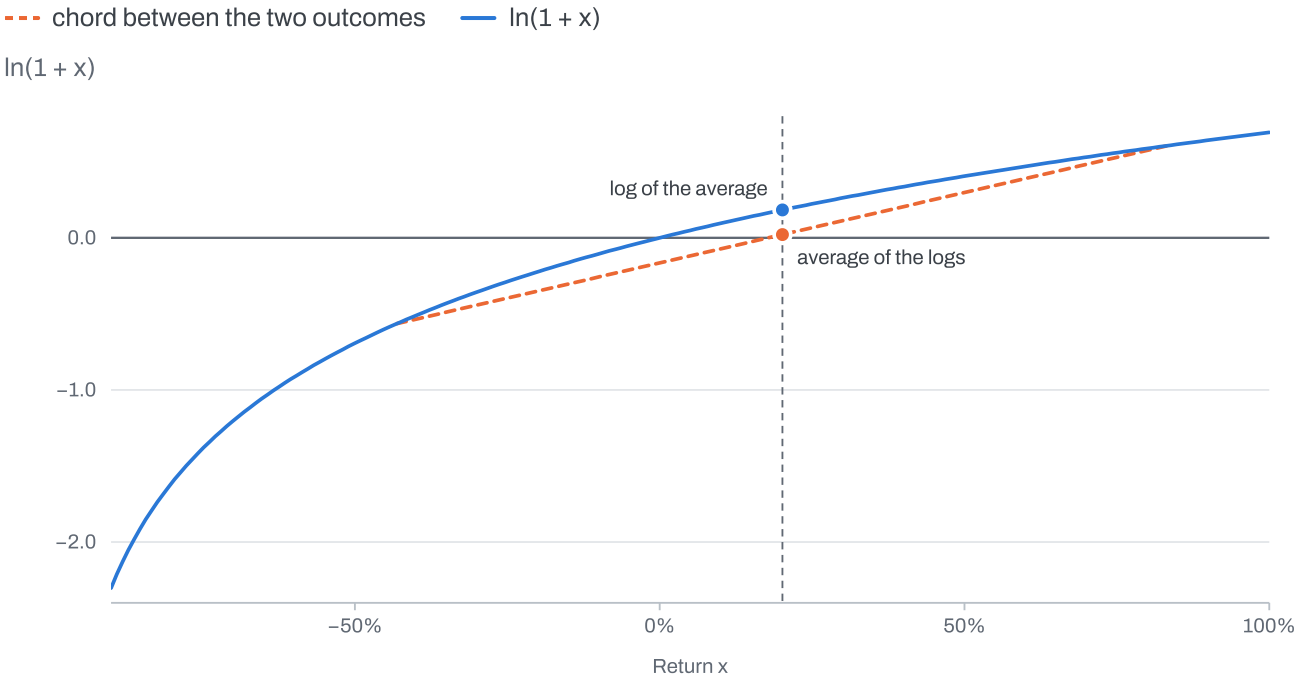
6 JENSEN'S INEQUALITY

- ☐ prove Jensen's inequality and its strict form from the tangent-line property, and derive the weighted AM–GM inequality from it

For a two-point return the logarithm's concavity is visible: the chord between the two outcomes on the curve $\ln(1 + x)$ lies below the curve. Above the mean μ, the chord gives the average of the logs, g, and the curve the log of the average, $\ln(1 + \mu)$. The widget starts from 2008 and 2009 with equal probability: $\mu = 20.1\%$, but $g = 2.141\%$ against $\ln(1 + \mu) = 18.345\%$.

Jensen's chord on a two-point return

INTERACTIVE



Mean μ · standard deviation σ

20.1% · 63.2%

Average of the logs g · log of the average ln(1 + μ)

2.141% · 18.345%

Compound rate R_G · (3.9)'s μ − σ²/2

2.16% · 0.16%

Jensen gap ln(1 + μ) − g · drag μ − R_G (points)

16.20 · 17.97

σ²/2 · second order σ²/(2(1 + μ)²) (points)

19.97 · 13.84

The return is +b with probability p and −a otherwise. The logarithm is concave, so the chord lies below the curve: above the mean μ, the chord gives the average of the logs, g, and the curve the log of the average, ln(1 + μ). Their difference is the Jensen gap; the volatility drag is μ − R_G with $R_G = e^g - 1$. Gaps are in percentage or log points.

THEOREM **Jensen's inequality**

Let h be concave and differentiable on an open interval $\mathcal{I}$, and let X be a random variable with

- values in $\mathcal{I}$ with probability one,
- finite expectations $\mathbb{E}|X|$ and $\mathbb{E}|h(X)|$.

Then

$$\mathbb{E}[h(X)] \leq h(\mathbb{E}[X]). \tag{14.10}$$

If h is strictly concave, equality holds only when $\Pr(X = \mu_X) = 1$. For a convex h both inequalities reverse.

Proof. The mean μ_X lies in $\mathcal{I}$. Indeed, X lies in the open interval $\mathcal{I}$ with probability one, so its distance to a finite end is positive with probability one. By (14.3), this distance has mean μ_X minus a lower end, or an upper end minus μ_X, at least 0 by monotonicity. It is not 0: if a variable $Y > 0$ with probability one had mean 0, Markov's (14.7) would give $\Pr(Y \geq x) = 0$ for every $x > 0$. Countable additivity would then give $\Pr(Y > 0) = 0$.

By LM4's Deep Dive on tangent lines, the concave h lies below its tangent at μ_X: for every $x \in \mathcal{I}$,

$$h(x) \leq h(\mu_X) + h'(\mu_X)(x - \mu_X).$$

Put $x = X$, which lies in $\mathcal{I}$ with probability one, and take expectations with (14.3) and monotonicity: $\mathbb{E}[h(X)] \leq h(\mu_X) + h'(\mu_X)(\mathbb{E}[X] - \mu_X) = h(\mu_X)$. ■

The proof needs only the tangent line, which covers every h in this book. For a convex h, apply (14.10) to $-h$: with $h(x) = e^x$ it gives $\mathbb{E}[e^X] \geq e^{\mathbb{E}X}$, and LM5's (5.12) measures the gap for a normal X, $\mathbb{E}[e^X] = e^{\mu_X + \sigma_X^2/2}$.

DEEP DIVE **The strict form** optional · click to expand

Here x_{mid} is a midpoint and ε a small positive number. Let h be strictly concave and differentiable on $\mathcal{I}$, and let $x \neq \mu_X$ lie in $\mathcal{I}$, with $x_{\text{mid}} := (x + \mu_X)/2$. Strict concavity puts $h(x_{\text{mid}})$ strictly above the chord, and the tangent at μ_X lies above h at x_{mid}:

$$\begin{aligned} \tfrac{1}{2}h(x) + \tfrac{1}{2}h(\mu_X) &< h(x_{\text{mid}}) \\ &\leq h(\mu_X) + h'(\mu_X)\,\frac{x - \mu_X}{2}. \end{aligned}$$

Doubling and subtracting $h(\mu_X)$ gives the strict tangent inequality $h(x) < h(\mu_X) + h'(\mu_X)(x - \mu_X)$ for every $x \neq \mu_X$.

So the tangent gap $h(\mu_X) + h'(\mu_X)(X - \mu_X) - h(X)$ is non-negative, positive when $X \neq \mu_X$, and has expectation $h(\mu_X) - \mathbb{E}[h(X)]$. If $\Pr(X \neq \mu_X) > 0$, countable additivity gives some $\varepsilon > 0$ for which the gap is at least ε with positive probability. Markov's (14.7) then makes its expectation at least ε times that probability, so (14.10) is strict. ■

PROPOSITION **Weighted AM–GM inequality**

Let $x_1, \ldots, x_J > 0$, and let the weights satisfy $w_j \geq 0$ and $\sum_j w_j = 1$. Then

$$\sum_j w_j x_j \geq \prod_j x_j^{w_j}, \tag{14.11}$$

with equality if and only if all x_j with $w_j > 0$ are equal.

Let X take the value x_j with probability w_j: the weights qualify as probabilities because they are non-negative and sum to 1. The logarithm is strictly concave on $\mathcal{I} = (0, \infty)$, so (14.10) gives $\sum_j w_j \ln x_j \leq$

$\ln \sum_j w_j x_j$; exponentiate. By the strict form, equality needs equal x_j wherever $w_j > 0$.

With $w_j = 1/J$ this is the strict unweighted inequality that LM1 used for its compounding sequence. With $x_j = e^{r_j}$ and logs, (14.11) is LM1's (1.14) for a long-only portfolio.

EXAMPLE 9

The case's first question

Apply (14.11) to Exhibit 1's 23 calendar-year growth factors with equal weights, and to 2008 and 2009 alone.

SOLUTION

With $w_j = 1/23$, the left side is one plus the arithmetic mean, 1.1497, and the right side one plus the compound rate, 1.1235. The compound rate cannot exceed the average, and with unequal factors it lies strictly below.

For 2008 and 2009 alone, $\frac{1}{2}(0.5693 + 1.8334) = 1.20135$, while $\sqrt{0.5693 \times 1.8334} = 1.02164$. The pair's compound rate is 2.16% a year against an average of 20.1%.

PITFALL **Check the direction and the weights**

Jensen's inequality reverses for a convex h, so check the sign of h'' first (LM4; Simon and Blume 1994, chapter 21). Its weights must be probabilities. LM1's weights $(1.5, -0.5)$, a short position, are not, and (1.14) fails for them; Problem 8 shows where the proof breaks.

導讀 **Jensen 不等式：切線的期望值**

凹函數（concave function）的圖形在每一條切線（tangent line）的下方。在平均數（mean）μ_X 取切線，代入 X 再取期望值（expectation），切線部分的期望值剛好等於 $h(\mu_X)$，於是得到 $\mathbb{E}[h(X)] \le h(\mathbb{E}X)$，這就是 Jensen 不等式（Jensen's inequality）；凸函數（convex function）的方向相反。條件要記清楚：X 必須以機率 1 落在 h 的開區間（open interval）內，而且兩個期望值都存在。把權重看成機率、取 $h = \ln$，就得到加權算幾不等式（weighted AM–GM inequality）。權重必須非負且總和為 1，放空造成的負權重不是機率，不等式就可能不成立。

KNOWLEDGE CHECK 5

Apply (14.10) to the convex function $h(x) = x^2$. What inequality results, and what is the difference between its two sides?

ANSWER

For a convex h the inequality reverses: $\mathbb{E}[X^2] \geq (\mathbb{E}X)^2$. By (14.4) the difference is $\mathbb{E}[X^2] - \mu_X^2 = \mathrm{Var}(X)$, which is zero only for a constant X.

7 VOLATILITY DRAG AND THE JENSEN GAP

- ☐ explain volatility drag through the Jensen gap $\ln(1+\mu) - g$ of the logarithm, compute both exactly, and judge when $\sigma^2/2$ describes them

Apply Jensen's inequality to the growth factor $1 + R$, with the logarithm as h.

RESULT The compound rate never exceeds the mean

Let $R > -1$ with probability one, with $\mathbb{E}|R|$ and $\mathbb{E}|\ln(1+R)|$ finite. Then

$$g \leq \ln(1+\mu), \qquad \text{so} \qquad R_G \leq \mu, \tag{14.12}$$

with equality if and only if $\Pr(R = \mu) = 1$.

Under $\hat{F}$, $g = \bar{r}$ and R_G is LM3's geometric mean return, so for any returns the compound rate never exceeds the average. The condition $R > -1$ matters: a normal R puts positive probability below −1, where $\ln(1+R)$ is undefined, so $\mathbb{E}[\ln(1+R)]$ does not exist and SciPy returns `nan`. LM4's quadratic approximation (4.2) hides this; a lognormal model, with $\ln(1+R)$ normal, keeps $R > -1$.

Two names for the gap

The **volatility drag** is LM3's $\mu - R_G$ ($\bar{R} - R_G$ in a sample), in percentage points; the **Jensen gap** is $\ln(1+\mu) - g$, in log points. Since $1 + R_G = e^g$,

$$\mu - R_G = (1+\mu)\left(1 - e^{-(\ln(1+\mu) - g)}\right),$$

so the drag is about $1 + \mu$ times the Jensen gap when both are small.

The Jensen gap has an exact series. Suppose that R takes finitely many values, each with $|R - \mu| < 1 + \mu$, and write

$$\ln(1+R) = \ln(1+\mu) + \ln\left(1 + \frac{R-\mu}{1+\mu}\right).$$

LM3's series (3.3) converges at every value of the last term, and taking expectations term by term, with $\mathbb{E}[R-\mu]=0$, gives

$$\ln(1+\mu)-g=\frac{\sigma^2}{2(1+\mu)^2}-\frac{\mathbb{E}[(R-\mu)^3]}{3(1+\mu)^3}+\frac{\mathbb{E}[(R-\mu)^4]}{4(1+\mu)^4}-\cdots \tag{14.13}$$

The third-order term is $-\text{Skew}\cdot\sigma^3/(3(1+\mu)^3)$, so negative skewness widens the gap. To second order in the returns, the factor $1/(1+\mu)^2$ is 1 and the gap is $\sigma^2/2$, the drag of LM3's (3.8) and (3.9).

For a lognormal law the gap needs no series. If $\ln(1+R)$ is normal with mean g and standard deviation σ_r, LM5's (5.12) gives $1+\mu=\mathbb{E}[e^{\ln(1+R)}]=e^{g+\sigma_r^2/2}$. The Jensen gap is then exactly $\sigma_r^2/2$, half the variance of the log return rather than of R.

EXAMPLE 10

The Jensen gap and the drag, exactly

Compute TAIEX's daily Jensen gap and drag, and split the gap into the terms of (14.13). Repeat for the calendar years and for 2008 and 2009, and compare the lognormal form.

SOLUTION

Daily, $\ln(1+\bar{R})-\bar{r}=0.7316$ bp. The second-, third- and fourth-order terms of (14.13) are $0.7292+0.0020+0.0004$ bp; negative skewness adds the third. The drag is $\bar{R}-R_G=(1+\bar{R})(1-e^{-0.7316\text{ bp}})=0.7320$ bp, against $s^2/2=0.7301$ bp (Exhibit 7).

For the calendar years, the Jensen gap is 2.31 log points, against 2.12 at second order and 2.30 through the fourth; the drag is 2.62 percentage points, against $\sigma^2/2=2.81$. For 2008 and 2009, the Jensen gap is 16.20, against 13.84 and 15.76, and the drag is 17.97, against 19.97. With moves this large the series converges slowly.

The lognormal form uses TAIEX's $s_r=1.2107\%$: $s_r^2/2=0.7329$ bp, against the Jensen gap of 0.7316 bp.

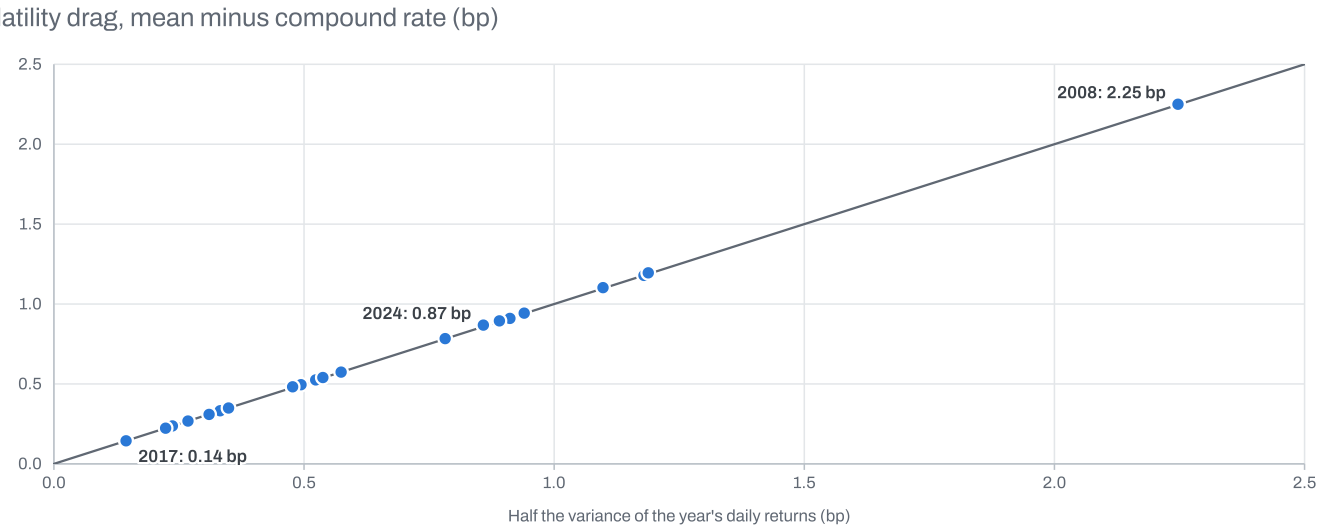
Exhibit 7: Volatility drag and Jensen gap of four laws

Law	μ	σ	Drag	$\sigma^2/2$	Jensen gap	Second order	To fourth order
TAIEX, daily (bp)	0.0625%	1.2084%	0.7320	0.7301	0.7316	0.7292	0.7316
TAIEX, years	14.97%	23.70%	2.62	2.81	2.31	2.12	2.30
2008 and 2009	20.1%	63.2%	17.97	19.97	16.20	13.84	15.76
+100% or −50%	25%	75%	25.00	28.125	22.31	18.00	21.24

Note: drag $\mu - R_G$ and $\sigma^2/2$ in percentage points; the Jensen gap $\ln(1+\mu) - g$, the term $\sigma^2/(2(1+\mu)^2)$ and (14.13) through the fourth order in log points; the daily row in bp. TAIEX rows: divisors T and 23, the years 2003–2025 (2003 from its first close, 2 January 2003). The pair takes the printed −43.07% and +83.34% with probability ½ each; two decimals of its μ and σ would be rounding ties. The last row is LM3's pitfall. Source: as Exhibit 1.

Within each calendar year of daily returns, half the variance describes the drag almost exactly (Exhibit 8).

Exhibit 8: Each calendar year's daily drag against half its variance



Note: one point per calendar year 2003–2025 (2003 from its first close, 2 January 2003), from that year's daily returns, in bp. Each point plots their volatility drag against half their variance, whose divisor is the year's count. The line is the 45° line. The largest difference, drag minus half the variance, is 0.0097 bp (2024), 1.13% of half the variance. The drag runs from 0.14 bp (2017) to 2.25 bp (2008). Source: as Exhibit 1.

Leverage enlarges the gap: applied to fR, (14.12) gives $g(f) \le \ln(1+f\mu)$, and the second-order term grows like f^2 (LM41). Which of the two rates an investor lives through over many periods is the subject of LM17.

PITFALL $\sigma^2/2$ **is a second-order term**

At daily frequency $\sigma^2/2$ misses TAIEX's drag by under 0.002 bp. For the calendar years, (3.9) gives $14.97\% - 23.70\%^2/2 = 12.16\%$ (12.17% from unrounded inputs), against the actual 12.35%; for 2008 and 2009, 0.16% against 2.16%. And (14.13) and (3.9) use the variance of R, the lognormal form that of r (0.7329 against 0.7301 bp).

導讀 **波動拖累的真正原因**

對 $1+R$ 套用 Jensen 不等式（Jensen's inequality），取 $h = \ln$，對數報酬（log return）的期望值（expectation）g 就不超過 $\ln(1+\mu)$，所以複利報酬率（compound rate）R_G 永遠不超過算術平均（arithmetic mean）μ。兩者的差距有兩種量法：波動拖累（volatility drag）$\mu - R_G$ 以百分點計，Jensen 差距（Jensen gap）$\ln(1+\mu) - g$ 以對數點（log points）計。把對數展開成級數（series），Jensen 差距的二階項是 $\sigma^2/(2(1+\mu)^2)$，三階項由偏態（skewness）決定，四階項由峰態（kurtosis）決定，LM3 的 $\sigma^2/2$ 只是二階近似。日報酬時二階項幾乎完全精準，年報酬或 2008、2009 年這種大波動時，高階項就不能忽略。若對數報酬服從常態分布（normal distribution），也就是對數常態分布（lognormal distribution）模型，Jensen 差距恰好等於對數報酬變異數（variance）的一半 $\sigma_r^2/2$。

KNOWLEDGE CHECK 6

A fund doubles TAIEX's daily return. Compare its daily Jensen gap, $\ln(1+2\bar{R}) - \mathbb{E}_{\hat{F}}[\ln(1+2R)]$, with four times the gap at $f = 1$.

ANSWER

The gap is 2.936 bp, against $4 \times 0.7316 = 2.926$ bp. The second-order term grows like f^2, and the higher-order terms, which grow faster, add the rest.

8 RANDOM VARIABLES AND EXPECTATIONS IN PYTHON

- [ ] compute the distribution, quantiles, moments and tail frequencies of an empirical distribution in Python, stating the quantile and moment conventions

A NumPy array of returns is its empirical distribution, provided each call uses the right convention (Exhibit 9).

Exhibit 9: NumPy and SciPy calls for distributions and moments

Call	Returns	Convention
`np.mean(R)`	$\bar{R} = \mathbb{E}_{\hat{F}}[R]$	
`np.var(R)` , `np.std(R)`	s^2, s	divisor T (`ddof=0` , the default)
`np.quantile(R, u, method="inverted_cdf")`	$\hat{F}^{-1}(u) = R_{(\lceil uT \rceil)}$	the quantile function of Section 2
`np.quantile(R, u)`	interpolates between order statistics	default `"linear"` : not $\hat{F}^{-1}$
`np.median(x)`	the middle value, or the midpoint of the middle pair	a fair die: 3.5
`stats.rv_discrete(values=(xk, pk))`	a discrete law with integer values `xk`	`.median()` and `.ppf(0.5)` of a die: 3
`stats.skew(R)` , `stats.kurtosis(R)`	skewness; excess kurtosis	divisor T (`bias=True`); add 3 for the kurtosis
`stats.norm(m, sd).moment(4)`	the raw moment $\mathbb{E}[X^4]$	(14.6)
`stats.norm(m, sd).expect(np.log1p)`	$\mathbb{E}[\ln(1+X)]$, by numerical integration	`nan` : a normal puts mass below −1

Note: `stats` is `scipy.stats` ; `m` and `sd` are the normal's mean and standard deviation. Checked with NumPy 2.5.3 and SciPy 1.18.1, the versions in `uv.lock` .

Every count against a threshold uses one guard, `R <= c + 1e-12` for $R \le c$ and `R >= c - 1e-12` for $R \ge c$. Index levels have two decimals, so a round return can be an exact tie that floating-point division puts on either side of the threshold. At −3% the guard and an exact count with fractions both give 108.

PYTHON

```python
import numpy as np
from scipy import stats

# R: TAIEX's 5,849 daily total returns as a NumPy array
Rbar, s = R.mean(), R.std()                      # divisor T
q = np.quantile(R, 0.001, method="inverted_cdf")  # R_(6), -6.313%
z = (R - Rbar) / s
lower4 = int((z <= -4 + 1e-12).sum())          # 17 days, with the guard
excess = stats.kurtosis(R)                       # 5.27, excess kurtosis
gap = np.log1p(Rbar) - np.log1p(R).mean()      # Jensen gap, 0.7316 bp
```

PITFALL `np.quantile`'s default interpolates

By default, `np.quantile` interpolates between order statistics and can return a value that is no day's return. At $u = 0.001$ it gives −5.866%, against $\hat{F}^{-1}(0.001) = -6.313\%$: the interpolation pulls a far-tail quantile towards the centre. Pass `method="inverted_cdf"` for the quantile function, and state the method with every quantile.

IN PYTHON Companion lab

The Polars-first notebook `LM14_lab.ipynb` rebuilds the module's exhibits and computational examples, works the computational practice problems, and ends with a stability check across the two price-limit regimes.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 7

Which call returns $\hat{F}^{-1}(0.001)$ for TAIEX, and what does the other call return?

ANSWER

`np.quantile(R, 0.001, method="inverted_cdf")` returns $R_{(6)} = -6.313\%$; the default returns −5.866%, an interpolated value.

SUMMARY

- A random variable maps outcomes to numbers. One distribution function serves discrete and continuous laws, the quantile function always exists, and the medians form a closed interval.
- Under its empirical distribution $\hat{F}$, a sample's frequencies are probabilities, $\hat{F}^{-1}(u) = R_{(\lceil uT \rceil)}$, and its divisor-$T$ statistics are population moments.
- Expectation (14.1) needs absolute convergence; LOTUS (14.2) and linearity (14.3) compute it.
- The identity $\mathbb{E}[X^2] = \sigma_X^2 + \mu_X^2$, (14.4), is (3.7) for any law; a normal has $\mathbb{E}[Z^4] = 3$, (14.6). TAIEX has skewness −0.34 and excess kurtosis 5.27.
- The bounds (14.7)–(14.9) hold exactly but loosely: 17 TAIEX days at $k = 4$ against 344.1.
- Jensen's inequality (14.10) follows from one tangent line and gives (14.11) and $R_G \le \mu$, (14.12).
- The Jensen gap $\ln(1+\mu) - g$ has the exact series (14.13). To second order in the returns, its leading term is LM3's $\sigma^2/2$: accurate for daily returns, not for annual ones. For a lognormal law the gap is $\sigma_r^2/2$.

PRACTICE PROBLEMS

1. Four independent days of LM4's bet at $f = 1$ (+2% or −1%, equal probability) give the growth factor $G = 1.02^{j}\,0.99^{4-j}$ after j up days. Its values 0.960596, 0.989705, 1.019696, 1.050596 and 1.082432 have probabilities 1/16, 4/16, 6/16, 4/16 and 1/16. Which statement is true, with medians as in Section 2?

   A. Its only median is 1.019696, below its mean 1.020151.

   B. Every value from 0.989705 to 1.019696 is a median.

   C. Its median is its mean, 1.020151.

2. For the G of Problem 1, tabulate F_G. Compute $\mathbb{E}[G]$, $\mathbb{E}[\ln G]$ by linearity and $\ln \mathbb{E}[G]$ to six significant figures, and from them the Jensen gap $\ln \mathbb{E}[G] - \mathbb{E}[\ln G]$ in bp. Explain the sign of the gap.

3. Which random variable has a finite expectation?

   A. A payoff of 2^{j} with probability 2^{-j}, for $j = 1, 2, \ldots$

   B. LM5's Cauchy variable, with density proportional to $1/(1 + x^{2})$

   C. A Pareto variable with density $1.5x^{-2.5}$ for $x \geq 1$

4. LM4's trade returns +1% with probability 0.9 and −5% otherwise. Compute μ, σ, $\mathbb{E}[R^{2}]$, $\mathbb{E}[R^{3}]$, the skewness and the kurtosis (as fractions where not round), and repeat at the exposure $f = 8$. Why do the skewness and the kurtosis not change, and what does an excess kurtosis near TAIEX's say about a two-point law?

5. Show $\mathbb{E}[Z^{6}] = 15$ by integration by parts, and deduce $\mathrm{Var}(Z^{3})$. For an illustrative $X \sim \mathcal{N}(0.05, 0.20^{2})$, compute $\mathbb{E}[X^{3}]$ and $\mathbb{E}[X^{4}]$ exactly with (14.6), and check them with `stats.norm(0.05, 0.2).moment`.

6. With TAIEX's printed $\bar{R} = 0.0625\%$ and $s = 1.2084\%$ (divisor T), bound the share of days with $R \leq -5\%$ by Cantelli's and by Chebyshev's inequality. Give k to four decimals and shares to two decimals of a percent. Compare with the 15 such days of LM13. Which bound is tighter, and why must both hold for $\hat{F}$?

7. For the Pareto variable of Problem 3, what does Chebyshev's inequality say about $\Pr(|X - \mu_X| \geq 3\sigma_X)$?

   A. It is at most 11.1%.

   B. Nothing, because the variance is infinite.

   C. It is at most 10%.

8. Prove (14.11) from (14.10), saying where each condition enters. Then take LM1's weights $(1.5, -0.5)$ and log returns $(0.10, 0.20)$: show where the proof fails, and compare $\ln(1.5e^{0.10} -$

$0.5e^{0.20}$) with $1.5 \times 0.10 - 0.5 \times 0.20$ to four decimals.

9. Which inequality holds for every positive X whose expectations below exist? Check each, to four decimals, on Problem 4's trade at its Kelly exposure: $X = 1 + 8R$ is 1.08 with probability 0.9 and 0.60 otherwise.

A. The inequality $\mathbb{E}[\ln X] \geq \ln \mathbb{E}[X]$

B. The inequality $\mathbb{E}[1/X] \leq 1/\mathbb{E}[X]$

C. The inequality $\mathbb{E}[\sqrt{X}] \leq \sqrt{\mathbb{E}[X]}$

10. TAIEX's calendar-year returns 2003–2025 (2003 from its first close) have $\mu = 14.97\%$, $\sigma = 23.70\%$ (divisor 23), skewness 0.19 and kurtosis 5.02. (a) Approximate the compound rate by (3.9). (b) Compute the three terms of (14.13) through the fourth order, their sum and the implied $R_G = (1+\mu)e^{-\text{gap}} - 1$. (c) The actual rate is 12.35%: which term matters most, and why? Give rates to two decimals of a percent, terms to four decimals of a log point.

11. *(Python)* Use the lab's TAIEX file, with divisor T. Give A to two decimals, rates to four decimals of a percent, quantiles to three, and the Jensen gap and its terms to four decimals of a bp. (a) Compute T, A, $\bar{R}$ and s, and check (3.7). (b) Compute $\hat{F}^{-1}(0.001)$ as $R_{(\lceil uT \rceil)}$ and with `np.quantile`'s default. (c) Recount Exhibit 6's days at $k = 2, 3, 4$ with the guard. (d) Compute the daily Jensen gap and the three terms of (14.13).

SOLUTIONS

1. **A** is correct. The distribution function is 5/16 just below 1.019696 and 11/16 at it, so $\Pr(G \leq 1.019696)$ and $\Pr(G \geq 1.019696)$ are both 11/16; no other value qualifies. The mean is $1.005^4 = 1.020151$ (Solution 2). B reads the flat stretch of F_G as medians, but $F_G = 5/16$ there. C fails because $\Pr(G \geq 1.020151) = 5/16$.

2. The distribution function is 1/16, 5/16, 11/16, 15/16 and 1 at the five values. By (14.1), the binomial theorem gives $\mathbb{E}[G] = ((1.02 + 0.99)/2)^4 = 1.005^4 = 1.02015$ (LM16 gives the product rule behind it). By linearity, $\mathbb{E}[\ln G] = 4 \times 0.487615\% = 1.95046\%$, while $\ln \mathbb{E}[G] = 4 \ln 1.005 = 1.99502\%$. The Jensen gap, 4.456 bp, is positive by (14.10): the logarithm is strictly concave and G is not constant.

3. **C** is correct: the mean is $\int_1^\infty x \cdot 1.5x^{-2.5}\,dx = 1.5/0.5 = 3$. In A every term $2^j \cdot 2^{-j}$ equals 1, so the sum diverges. In B the integral of $|x|/(1+x^2)$ diverges like a logarithm.

4. The mean is $\mu = 0.9 \times 1\% - 0.1 \times 5\% = 0.4\%$. The deviations are +0.6% and −5.4%, so $\sigma^2 = 0.9 \times 0.36 + 0.1 \times 29.16 = 3.24$ (%²) and $\sigma = 1.8\%$. Then $\mathbb{E}[R^2] = 3.4 \times 10^{-4} = \sigma^2 + \mu^2$ and $\mathbb{E}[R^3] = 0.9 \times 10^{-6} - 0.1 \times 1.25 \times 10^{-4} = -1.16 \times 10^{-5}$.

   The standardized values are $+\frac{1}{3}$ (probability 0.9) and -3, so the skewness is $0.9/27 - 0.1 \times 27 = -8/3 = -2.67$ and the kurtosis $0.9/81 + 0.1 \times 81 = 73/9 = 8.11$, an excess of 5.11. At $f = 8$ the return is $8R$, with $\mu = 3.2\%$, $\sigma = 14.4\%$, $\mathbb{E}[(8R)^2] = 64 \times 3.4 \times 10^{-4} = 0.02176$, which is $0.144^2 + 0.032^2$, and $\mathbb{E}[(8R)^3] = 512 \times (-1.16 \times 10^{-5}) = -0.0059392$. The standardized values do not change, so by (14.5) neither do the skewness and kurtosis.

The excess kurtosis is close to TAIEX's 5.27, yet this law has no tail: kurtosis measures how much of the variance comes from rare large deviations, here one rare loss. One number cannot tell a fat tail from a rare jump.

5. Moving one factor z onto φ as in Section 4, $\mathbb{E}[Z^6] = \big[-z^5\varphi(z)\big]_{-\infty}^{\infty} + 5\mathbb{E}[Z^4] = 15$, and $\operatorname{Var}(Z^3) = \mathbb{E}[Z^6] - 0^2 = 15$. With $\mu = 0.05$ and $\sigma^2 = 0.04$, (14.6) gives $\mathbb{E}[X^3] = 0.000125 + 0.006 = 0.006125$ and $\mathbb{E}[X^4] = 0.00000625 + 0.0006 + 0.0048 = 0.00540625$, the values SciPy returns.

6. Here $k = (0.0625 + 5)/1.2084 = 4.1894$, so Cantelli's bound is $1/(1 + k^2) = 5.39\%$. Chebyshev's two-sided event contains the lower tail, so $1/k^2 = 5.70\%$ bounds it too, less tightly. The data have 15 days, 0.26%. Both hold because $\hat{F}$ is a probability law with mean $\bar{R}$ and variance s^2.

7. **B** is correct. The integral $\int_1^\infty x^2 \cdot 1.5x^{-2.5}\,dx$ diverges, so the variance is infinite and (14.8) does not apply. A uses $1/3^2$ without that condition. C is Cantelli's $1/(1 + 3^2)$, which needs a finite variance too and bounds one tail only.

8. Let X take the value x_j with probability w_j, which needs non-negative weights summing to 1. Positive x_j put X in $\mathcal{I} = (0, \infty)$, where the logarithm is strictly concave and differentiable; finitely many values make both expectations finite. Then (14.10) gives $\sum_j w_j \ln x_j \le \ln \sum_j w_j x_j$; exponentiate, and use the strict form for equality.

   With weights $(1.5, -0.5)$ one probability would be −0.5, so no such X exists. Indeed, $1.5e^{0.10} - 0.5e^{0.20} = 1.0471$, whose logarithm, 0.0460, lies below 0.0500.

9. **C** is correct. The square root is concave, so (14.10) gives $\mathbb{E}[\sqrt{X}] \le \sqrt{\mathbb{E}[X]}$: 1.0128 against 1.0159. A reverses (14.10) for the logarithm: $\mathbb{E}[\ln X] = 1.82\%$ against $\ln \mathbb{E}[X] = 3.15\%$. B reverses it for the convex $1/x$: $\mathbb{E}[1/X] = 1$ exceeds $1/\mathbb{E}[X] = 0.9690$. Here $\mathbb{E}[1/X] = 1$ exactly, because LM4's Kelly condition $\mathbb{E}[R/(1 + fR)] = 0$ gives $\mathbb{E}[1/(1 + fR)] = 1$.

10. (a) By (3.9), $R_G \approx 14.97\% - 23.70\%^2/2 = 12.16\%$. (b) The terms are $0.2370^2/(2 \times 1.1497^2) = 2.1247$, $-0.19 \times 0.2370^3/(3 \times 1.1497^3) = -0.0555$ and $5.02 \times 0.2370^4/(4 \times 1.1497^4) = 0.2266$ log points, with sum 2.2958, so $R_G = 1.1497\, e^{-0.022958} - 1 = 12.36\%$.

    (c) The second-order term carries most of the gap, but alone it gives $R_G = 12.55\%$. Adding the fourth-order term, large because the kurtosis is 5.02, gives 12.30%. The third-order term is small because the skewness is, and it raises the estimate to 12.36%.

    (3.9) misses for another reason: it subtracts $\sigma^2/2 = 2.81$ points as the drag. The drag is about $1 + \mu$ times the gap, so its leading term is $\sigma^2/(2(1 + \mu)) = 2.44$ points, and the higher terms bring it to the actual 2.62.

11. (a) The sample has $T = 5{,}849$ returns, $A = 246.12$, $\bar{R} = 0.0625\%$ and $s = 1.2084\%$; (3.7) reads $1.464 \times 10^{-4} = 1.460 \times 10^{-4} + 3.907 \times 10^{-7}$. (b) The quantile is $R_{(6)} = -6.313\%$; the default gives −5.866%. (c) Two-sided, 319, 99 and 30 days; lower, 188, 61 and 17. (d) The Jensen gap is 0.7316 bp, with terms $0.7292 + 0.0020 + 0.0004$ bp.

GLOSSARY

Term	中文	Meaning
Random variable	隨機變數	A function that assigns a number to each outcome of a sample space.
Distribution (law)	分布	The probabilities with which a random variable takes its values, given by p_X or f_X and by F_X.
Probability mass function	機率質量函數	$p_X(x_j) = \Pr(X = x_j)$ for a discrete X.
Distribution function	累積分布函數	$F_X(x) = \Pr(X \le x)$, for discrete and continuous X alike.
Quantile function	分位數函數	$F_X^{-1}(u) = \min\{x : F_X(x) \ge u\}$, $0 < u < 1$.
Median	中位數	A value x with $\Pr(X \le x) \ge \frac{1}{2}$ and $\Pr(X \ge x) \ge \frac{1}{2}$; the medians form a closed interval.
Empirical distribution	經驗分布	$\hat{F}$: mass $1/T$ on each observation of a sample.
Order statistic	順序統計量	$R_{(j)}$, the observation of rank j from the smallest; $\hat{F}^{-1}(u) = R_{(\lceil uT \rceil)}$.
Population; sample	母體；樣本	A property of a random variable or model; a statistic of data, with its divisor stated.
Expectation	期望值	$\mathbb{E}[X] = \sum_j x_j p_X(x_j)$, (14.1), or (5.9) for a density, when it converges absolutely.
Indicator	指示函數	$\mathbf{1}_{\mathcal{A}}$: 1 on the event $\mathcal{A}$, 0 elsewhere; $\mathbb{E}[\mathbf{1}_{\mathcal{A}}] = \Pr(\mathcal{A})$.
LOTUS	LOTUS	$\mathbb{E}[h(X)] = \sum_j h(x_j) p_X(x_j)$, (14.2).
Variance	變異數	$\mathrm{Var}(X) = \mathbb{E}[(X - \mu_X)^2] = \mathbb{E}[X^2] - \mu_X^2$, (14.4).
Skewness; kurtosis	偏態；峰態	Third and fourth standardized moments, (14.5).
Excess kurtosis	超額峰態	Kurtosis minus 3, the normal's value.
Markov's inequality	馬可夫不等式	$\Pr(X \ge x) \le \mathbb{E}[X]/x$ for $X \ge 0$ and $x > 0$, (14.7).
Chebyshev's inequality	謝比雪夫不等式	$\Pr(\lvert X - \mu_X \rvert \ge k\sigma_X) \le 1/k^2$, (14.8).
Cantelli's inequality	Cantelli 不等式	$\Pr(X - \mu_X \le -k\sigma_X) \le 1/(1 + k^2)$, (14.9).
Jensen's inequality	Jensen 不等式	$\mathbb{E}[h(X)] \le h(\mathbb{E}X)$ for a concave h, (14.10); reversed for a convex h.
Weighted AM–GM inequality	加權算幾不等式	$\sum_j w_j x_j \ge \prod_j x_j^{w_j}$ for $x_j > 0$, $w_j \ge 0$ and $\sum_j w_j = 1$, (14.11).
Compound rate	複利報酬率	$R_G = e^g - 1$, the constant rate with the same growth, LM3's geometric mean return (幾何平均報

Term	中文	Meaning
		酬); $R_G \le \mu$, (14.12).
Volatility drag	波動拖累	$\mu - R_G$ ($\bar{R} - R_G$ in a sample), in percentage points.
Jensen gap	Jensen 差距	$\ln(1+\mu) - g$, in log points, with the series (14.13).

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